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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

A Project Report On

Air Quality Index Prediction and Data Analysis

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Abstract

Air pollution is a critical global issue that significantly impacts human health and the environment. Monitoring air quality is essential for mitigating its adverse effects and implementing effective policies. This project focuses on developing an Air Quality Index (AQI) prediction system using machine learning techniques. The model utilizes air pollutants data collected from aqiweather.com website which includes particulate matter (PM2.5 and PM10), nitrogen dioxide (NO2), sulfur dioxide (SO2), ground level ozone (O3) and carbon monoxide, to predict AQI levels. Various data preprocessing techniques and machine learning algorithms have been employed to enhance prediction accuracy. The study provides insights into the performance of different models and identifies the most effective approach for AQI forecasting. The results demonstrate the potential of data-driven methods in air quality monitoring, enabling timely interventions and improved public health strategies.

Keywords: Air Quality Index, Machine Learning, Pollution Monitoring, Evaluation Matrix



GitHub URL: <https://github.com/shuvra24/AirQualityIndexandDataAnalysis>

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1.1 Introduction

1.1.1 Problem Statement

Air pollution has become one of the most pressing environmental concerns worldwide. The increasing concentration of pollutants in the atmosphere poses severe health risks, particularly in urban areas. Exposure to high levels of air pollution is linked to respiratory diseases, cardiovascular problems, and premature mortality. Despite existing monitoring systems, real-time AQI prediction remains a challenge due to the dynamic nature of air pollutants. Traditional monitoring methods often rely on static data collection, limiting their ability to provide accurate forecasts. The need for a robust, data-driven AQI prediction model has become imperative to enhance public awareness and inform policy decisions.

1.1.2 Objective

Chattogram is one of the fastest growing cities of Bangladesh with population 5.514 million (2024). Unplanned urbanization and growing industrial sector are contributing to air pollution in Chattogram city which is causing serious health issues to a larger resident. Our project will basically depend on real time pollutants level to calculate Air Quality Index in the industrial area of Chattogram. Analysis of data will help to identify the most polluted area and contributing factors to air pollution. So that, proactive measures can be taken to minimize pollution and also creating awareness at the individual level.

1.1.3 Motivation

Cities of Bangladesh are often in the headlines for breaking records in air pollution. Recent study has shown that pollutants concentration in Bangladesh is 15 times higher than the WHO guideline. This situation is comparatively worse in cities. In the recent time, deaths registered that are being caused for respiratory disease can't be a coincidence. This is an emergency situation and an environment crisis which is not addressed properly, which is happening most probably because it is an invisible problem and lack of monitoring stations. Even though, it's now not a matter of concern among us, data can be used to represent completely different scenario. By using machine learning approach, analysis of data in this respect, can help to address the concern about air pollution in the business capital of Bangladesh so that the people and all other living beings can have a healthier environment to live.

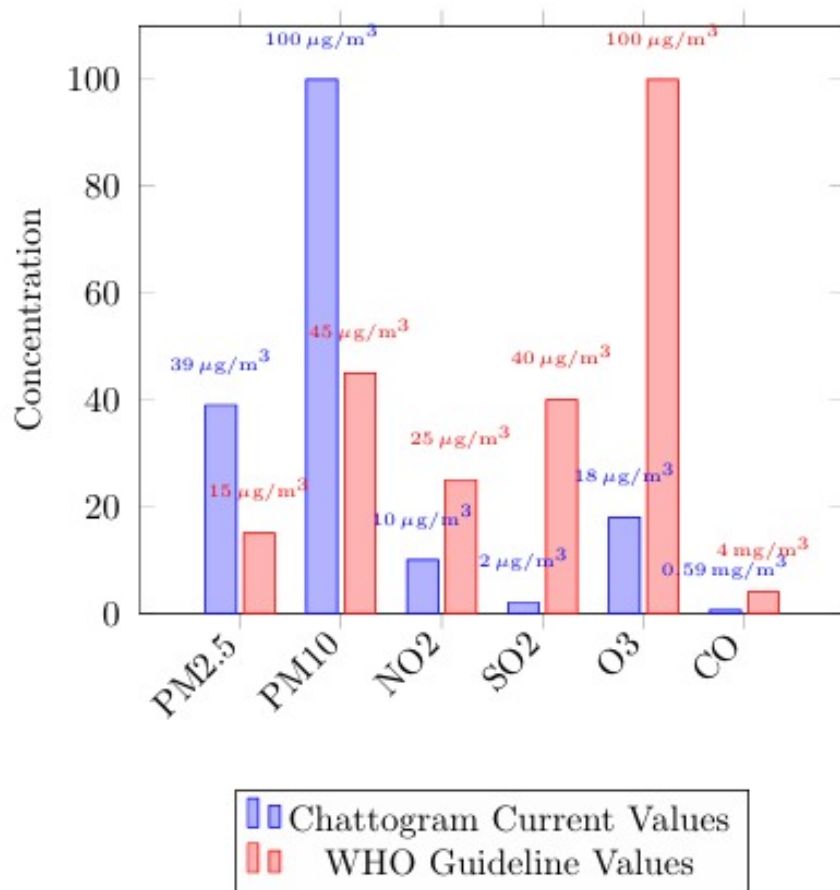


Figure 1.1: Comparison of Air Pollutant Levels in Chattogram(average) vs WHO Global Air Quality Guidelines 2021(maximum safe levels)

1.1.4 Contributions

The key contributions of this project are as follows:

- * Development of a machine learning based AQI prediction model
- * Implementation of data preprocessing techniques to improve model performance.
- * Evaluation and comparison of different machine learning algorithms for AQI prediction.
- * Analysis of the impact of various pollutants on AQI levels.

1.1.5 Report Outline

This report is structured as follows:

Chapter 2 presents a literature review on air quality monitoring and machine learning techniques used in AQI prediction.

Chapter 3 discusses the methodology, including data collection, preprocessing, and model development.

Chapter 4 provides experimental results, model evaluation, and performance analysis.

Chapter 5 concludes the study with key findings and suggests future improvements. This structured approach ensures a comprehensive understanding of AQI prediction and its applications in environmental health management.

2.1 LITERATURE REVIEW

2.1.1 Overview of the topic

Air pollution has been a subject of extensive research due to its impact on human health and the environment. Traditional air quality monitoring methods rely on ground-based sensors, which, while accurate, are limited in coverage and often require significant resources. With the advent of machine learning and data-driven techniques, predictive models have gained attention for their ability to forecast AQI based on historical and real-time data. This chapter reviews existing literature on AQI prediction, highlighting the methodologies and advancements in this field. Air quality prediction is crucial for protecting public health and managing urban pollution, especially in rapidly growing cities. ML techniques have become popular due to their ability to handle complex, non-linear relationships in environmental data. The reviewed studies focus on predicting the Air Quality Index (AQI) or specific pollutants like PM_{2.5} and NO_x, using data from various regions like Malaysia, Taiwan, India, and China.

2.1.2 Key Points:

- Research suggests machine learning (ML) is effective for predicting air quality, with various algorithms like Support Vector Machines (SVM), Random Forest, and Neural Networks showing promise.
- It seems likely that ensemble methods, such as Stacking and AdaBoost, often outperform individual models, though performance varies by region and data quality.
- The evidence leans toward deep learning techniques, like LSTM and MLP, being increasingly used, especially for capturing temporal patterns in air quality data.
- Challenges include data scarcity, long-term prediction difficulties, and the need for standardized evaluation, with ongoing research aiming to address these issues.

2.1.3 Methodologies used:

Different studies employ a range of ML methods, with SVM and Random Forest often used for short-term predictions, while deep learning excels in longer-term forecasts. For example, one study found SVM with an RBF kernel achieved high accuracy ($R^2 = 0.9843$) for API prediction in Malaysia. Another highlighted that Decision Trees performed best for AQI in Pune, India, especially on cloud platforms, reducing execution time significantly.

An unexpected detail is the impact of external factors, like the 2020 COVID-19 lockdown in India, which showed a sharp decline in pollutants, affecting ML model performance and highlighting the need for adaptive models.

The reviewed studies were selected based on their focus on ML techniques for air quality prediction, covering a range of algorithms such as Support Vector Machines (SVM), Random Forest, Neural Networks, Ensemble Methods, and Decision Trees. Data sources include government monitoring stations, sensor networks, and meteorological data, with preprocessing steps like handling missing values, normalizing data, and addressing outliers being common. The studies evaluate model performance using metrics such as Root Mean Square Error (RMSE), Mean Absolute Error (MAE), and Coefficient of Determination (R^2), with a focus on short-term predictions (1–24 hours) and specific pollutants like PM_{2.5}, NO_x, and CO.

2.1.4 Detailed Analysis of Studies:

1. Leong, W. C., Kelani, R. O., and Ahmad, Z. (2019) - Prediction of Air Pollution Index (API) Using Support Vector Machine (SVM)

This study, conducted in Malaysia, proposes using SVM with different kernel functions to predict the Air Pollution Index (API), which is crucial for assessing air quality and its health impacts. Data from 2009 to 2014, collected from Perak and Penang, were preprocessed to handle missing values and outliers using the interquartile range (IQR) rule. The dataset was split into 70% training, 15% validation, and 15% testing sets. Results showed that the RBF kernel outperformed others, achieving an SSE of 2008, MSSE of 1.444, and R^2 of 0.9843, indicating high accuracy. The study concludes that SVM, particularly with RBF, offers a simpler and more accurate alternative to traditional API calculation methods, emphasizing the importance of outlier removal and kernel selection.

2. Méndez, M., Merayo, M. G., and Núñez, M. (2023) - Machine Learning Algorithms to Forecast Air Quality: A Survey

Published in Artificial Intelligence Review, this survey reviews 155 papers from 2011 to 2021, focusing on ML, particularly deep learning (DL), for air quality forecasting. It highlights geographical trends, with China leading due to pollution severity and research capacity, and notes a steady increase in publications until 2020, with a decline in 2021 due to COVID-19-related studies. Predicted measures include AQI and PM_{2.5}, with DL algorithms like LSTM and MLP being more prevalent and effective than regression methods like SVR and Random Forest. The study emphasizes combining pollutant and weather features for better accuracy, with evaluation metrics like RMSE and MAE being common, and suggests future trends like Deep Transformer Networks.

3. Kang, G. K., Gao, J. Z., Chiao, S., Lu, S., and Xie, G. (2018) - Air Quality Prediction Using Big Data and Machine Learning Approaches

This paper, published in the International Journal of Environmental Science and Development, discusses the use of big data and ML for air quality prediction, focusing on six criteria pollutants (CO, Pb, NO₂, O₃, PM, SO) in the U.S. context. It reviews techniques like Artificial Neural Networks (ANN), Genetic Algorithm-ANN (GA-ANN), Random Forest, Decision Tree, Least Squares Support Vector Machine (LS-SVM), and Deep Belief Network. Random Forest achieved 81% accuracy for AQI in Shenyang, China, while LS-SVM showed promise for meteorological and pollution forecasting. The study identifies challenges like data quality issues and the need for real-time monitoring systems, suggesting future research in big data quality assurance and dynamic modeling.

4. Al-Eidi, S., Amsaad, F., Darwish, O., Tashtoush, Y., Alqahtani, A., and Niveshitha, N. (2023) - Comparative Analysis Study for Air Quality Prediction in Smart Cities Using Regression Techniques

Published in IEEE Access, this study compares Decision Tree, Linear, and Random Forest regression models for AQI prediction in Pune City, India, using a 2019 dataset of 103,205 records. Data preprocessing involved handling missing values and outliers, with feature selection using correlation analysis and data balancing using SMOTER. Results showed Decision Tree outperforming others, with lower MAE and RMSE, especially on cloud platforms, reducing execution time by 39% compared to local execution. The study highlights the benefits of cloud computing for scalability and recommends future research incorporating meteorological data.

5. Chen, S.-J., et al. (2020) - Machine Learning-Based Prediction of Air Quality

This Applied Sciences paper uses ML techniques to predict AQI in Taiwan based on an 11-year dataset (2008–2018). It evaluates Adaptive Boosting (AdaBoost), ANN, Random Forest, Stacking Ensemble, and SVM for 1, 8, and 24-hour predictions across three regions: Zhongli, Changhua, and Fengshan. Stacking Ensemble consistently outperformed others in R^2 and RMSE, while AdaBoost provided the best MAE. Prediction accuracy decreased with longer time frames, with Fengshan showing the best performance due to less decay over time. The study developed a forecasting system using Stacking Ensemble and AdaBoost, providing 95% confidence intervals.

6. Iskandaryan, D., Ramos, F., Trilles, S. (2020) - Air Quality Prediction in Smart Cities Using Machine Learning Technologies Based on Sensor Data: A Review

Another Applied Sciences review, this paper examines 41 studies from 2002 onwards, focusing on ML for air quality prediction in smart cities using sensor data. Neural networks were the most used (38%), followed by regression (24%) and ensemble methods (22%), with PM_{2.5} being the main pollutant targeted (19 studies). China was the primary case study country, with 66% of studies using hourly data and 49% using open data. Challenges include long-term prediction difficulties and high computational time

for neural networks, with future work suggesting standardized comparisons and exploring spatial attention.

7. Kumar, K., Pande, B. P. (2023) - Air Pollution Prediction with Machine Learning: A Case Study of Indian Cities

Published in the International Journal of Environmental Science and Technology, this study analyzes air quality in 23 Indian cities over six years (2015–2020), using ML models like KNN, Gaussian Naive Bayes, SVM, Random Forest, and XGBoost. Data preprocessing involved median imputation and logarithmic transformations, with XGBoost outperforming others in accuracy, precision, recall, and error metrics (MAE, RMSE, RMSLE, R^2). The 2020 COVID-19 lockdown significantly reduced pollutants, affecting model performance, and the study fills a research gap by focusing on Indian cities, suggesting future exploration of deep learning.

8. Liu, H., Li, Q., Yu, D., and Gu, Y. (2019) - Air Quality Index and Air Pollutant Concentration Prediction Based on Machine Learning Algorithms

This Applied Sciences paper uses SVR and Random Forest Regression (RFR) to predict AQI in Beijing and NOx concentrations in an Italian city, based on datasets from 2013–2018 and 2004–2005, respectively. SVR was more effective for AQI prediction ($RMSE = 7.666$, $R^2 = 0.9776$), while RFR performed better for NOx ($RMSE = 83.6716$, $R^2 = 0.8401$). The study explored pollutant correlations, concluding that RFR is generally superior due to lower time complexity, especially with larger datasets, providing insights for urban air quality modeling.

2.1.5 Comparative Analysis:

To facilitate comparison, the following table summarizes key aspects of the studies, including the ML techniques used, prediction targets, and performance metrics: Study ML Techniques Prediction Target Performance Metrics Region

This table highlights the diversity of ML techniques and the variability in performance metrics, with ensemble methods and deep learning often showing superior results.

Study	ML Techniques	Prediction Target	Performance Metrics	Region
Leong et al. (2019)	SVM (RBF kernel)	API	SSE, MSSE, R^2 (0.9843)	Malaysia
Méndez et al. (2023)	DL (LSTM, MLP), Regression	AQI, PM2.5	RMSE, MAE	Global (Survey)
Kang et al. (2018)	ANN, GA-ANN, Random Forest, LS-SVM	AQI, Pollutants	RMSE, Accuracy (81%)	U.S., China
Al-Eidi et al. (2023)	Decision Tree, Linear, Random Forest	AQI	MAE, RMSE, R^2	India (Pune)
Chen et al. (2020)	AdaBoost, ANN, Random Forest, Ensemble, SVM	AQI	RMSE, MAE	Taiwan
Iskandaryan et al. (2020)	Neural Networks, Regression, Ensemble	PM2.5, Pollutants	RMSE, MAE, R^2	Global (Survey)
Kumar and Pande (2023)	KNN, Naive Bayes, SVM, Random Forest, XGBoost	AQI	MAE, RMSE, R^2	India (23 cities)
Liu et al. (2019)	SVR, Random Forest Regression	AQI, NOx	RMSE, R^2	China, Italy

Table 2.1: Comparison of ML techniques for air quality prediction

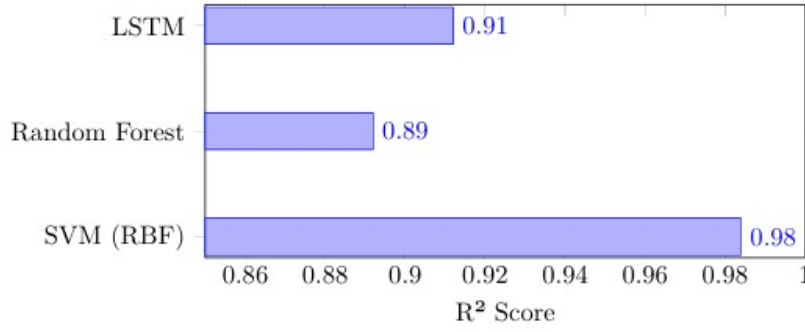


Figure 2.1: Comparative R² Scores from Literature

2.1.6 Key Findings and Trends

- Algorithm Performance:** No single algorithm is universally superior, with SVM and Random Forest effective for short-term predictions, and deep learning methods like LSTM and MLP excelling in capturing temporal patterns. Ensemble methods, such as Stacking and AdaBoost, often outperform individual models, as seen in Chen et al. (2020) and Kumar and Pande (2023), where XGBoost achieved the highest accuracy.
- Regional Variations:** Prediction accuracy varies by region due to differences in pollution levels and meteorological conditions. For example, Chen et al. (2020) found better performance in Taiwan’s southern region (Fengshan), while Kumar and Pande (2023) noted severe pollution in Indian cities like Delhi and Patna, affecting model outcomes.
- Data Quality and Integration:** High-quality, comprehensive datasets are crucial, with studies like Iskandaryan et al. (2020) emphasizing the increasing use of open data (49% of reviewed papers) and the integration of weather and spatial features for improved accuracy. Data preprocessing, such as outlier removal and balancing, significantly enhances model performance, as shown by Leong et al. (2019) and Al-Eidi et al. (2023).
- Temporal Considerations:** Most studies focus on short-term predictions (1–24 hours), with hourly data being common. Long-term predictions remain challenging, with accuracy decreasing over longer time frames, as noted by Chen et al. (2020) and Iskandaryan et al. (2020), suggesting a need for advanced models like deep transformers.
- Impact of External Factors:** An unexpected finding is the impact of external events, such as the 2020 COVID-19 lockdown, which Kumar and Pande (2023) observed led to a sharp decline in pollutants in Indian cities, affecting ML model performance and highlighting the need for adaptive models.

2.1.7 Scope

Air quality prediction has emerged as a critical research area due to the escalating health and environmental impacts of air pollution, particularly in urban and industrial settings. With urbanization projected to see 68% of the global population living in cities by 2050, the need for accurate and timely air quality forecasting is paramount. Machine learning (ML) techniques have gained traction for their ability to model complex, non-linear relationships in environmental data, offering a promising alternative to traditional statistical methods. This survey note synthesizes eight key studies published between 2018 and 2023, focusing on the application of ML for predicting air quality indices (AQI) and pollutant concentrations across diverse geographical regions, including Malaysia, Taiwan, India, China, and Europe. Chattogram is one of the largest growing cities in the world. It has immense possibilities to become the economic hub of south Asia. To achieve sustainable development goals, ensuring a healthier environment to reside with is much needed. The review aims to highlight methodologies, key findings, challenges, and future directions, providing a comprehensive overview for researchers and policymakers and using them to ensure cleaner air for the residents of Chattogram.

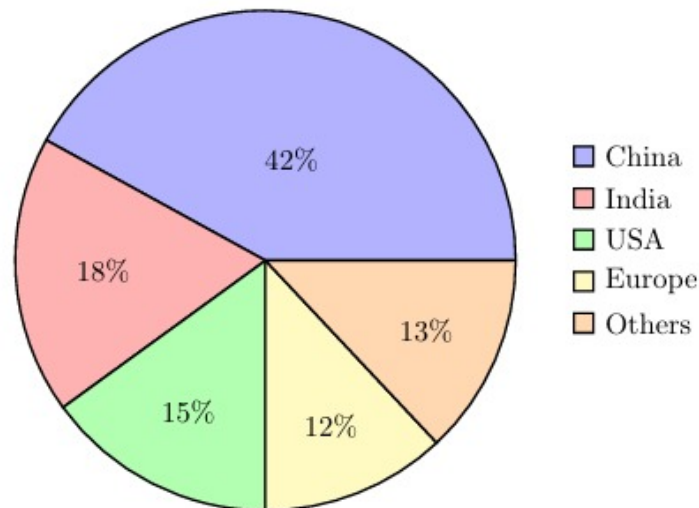


Figure 2.2: Geographical Distribution of Reviewed Studies

2.1.8 Challenges and Future Directions:

Several challenges persist, including data scarcity, especially for long-term predictions, and high computational time for some models like neural networks, as noted by Iskandaryan et al. (2020). Data quality issues, such as sensor faults and communication problems, were highlighted by Kang et al. (2018), suggesting a need for research in big data quality assurance. Future work should focus on standardized evaluation frameworks to compare results across studies, as suggested by Iskandaryan et al. (2020), and explore emerging techniques like Graph Neural Networks and Temporal Convolutional Networks, as proposed by Méndez et al. (2023).

2.1.9 Conclusion of the review:

The reviewed literature underscores the significant potential of machine learning in air quality prediction, offering tools to address the growing challenges of urban pollution. SVM, Random Forest, and deep learning methods are particularly effective, with ensemble techniques often providing the best results. However, the choice of algorithm depends on the specific context, including data availability, prediction horizon, and regional characteristics. The integration of diverse data sources, improved long-term forecasting, and standardized evaluation are critical areas for future research, ensuring ML can effectively support public health and environmental decision-making in smart cities.

3.1 Methodology

This section outlines the methodology employed for developing an AI-based Air Quality Index (AQI) prediction model using air quality measurements collected in Chattogram city, Bangladesh. The approach encompasses data collection, preprocessing, model training, and evaluation using supervised machine learning techniques to analyze and predict pollution trends.

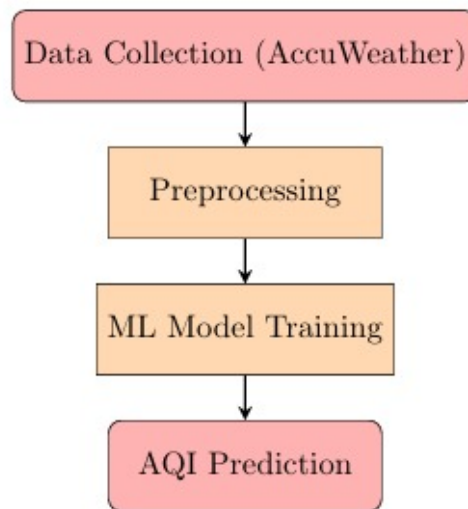


Figure 3.1: Workflow of AQI Prediction System

3.1.1 Data Collection

Air quality data was manually collected from a mobile weather application (AccuWeather.com) at five distinct locations in Chattogram city: Dewanhat, Anderkilla, Kotowali, Bayejid, and New Market. The dataset includes key air quality parameters—carbon monoxide

(CO), sulfur dioxide (SO), nitrogen dioxide (NO), ozone (O), particulate matter (PM_{2.5}), and particulate matter (PM₁₀)—measured in micrograms per cubic meter ($\mu\text{g}/\text{m}^3$). Measurements were recorded at four fixed time intervals daily: 12:00 AM, 6:00 AM, 12:00 PM, and 6:00 PM. This dataset serves as the foundation for analyzing air quality trends and developing predictive models for AQI and its associated categories.

3.1.2 Data Preprocessing

The collected data underwent thorough preprocessing to ensure suitability for machine learning tasks. The preprocessing steps included:

- **Data Cleaning:** Handling missing values by imputation or removal, and addressing outliers to maintain data integrity.
- **Normalization:** Scaling pollutant concentration values to a standardized range to ensure consistent model training.
- **AQI Calculation:** Computing AQI values based on pollutant concentration breakpoints as per standard air quality guidelines. The dataset was structured for two tasks:
 - **Regression:** Predicting continuous AQI values.
 - **Classification:** Predicting discrete AQI categories (e.g., Good, Moderate, Unhealthy).
- **Feature Engineering:** Structuring the dataset with pollutant concentrations (CO, SO, NO, O, PM_{2.5}, PM₁₀) as input features and AQI values or categories as target labels.

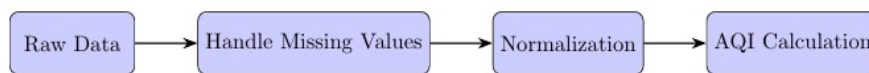


Figure 3.2: Data Preprocessing Steps

3.1.3 Model Training

Supervised learning algorithms were employed due to the availability of labeled data, which included pollutant concentrations and corresponding AQI values or categories. The following algorithms were selected for their suitability in handling regression and classification tasks:

- **Regression Models (for predicting continuous AQI values):**

- Linear Regression
- Decision Tree Regressor
- Random Forest Regressor

- **Classification Models (for predicting AQI categories):**

- Support Vector Machine (SVM)
- Naive Bayes Classifier
- Multilayer Perceptron (MLP)

The rationale for using supervised learning includes the availability of labeled data, the need to predict continuous values (regression) or discrete categories (classification), goal-oriented learning, and the ability to quantify performance using standard metrics. The dataset was split into training and testing sets (80/20 ratio) to facilitate model training and evaluation. During training, the models learned to map input features (pollutant concentrations) to output labels (AQI values or categories).

3.1.4 Model Architecture:

Linear Regression Model:

Linear Regression is a simple yet effective model for predicting the Air Quality Index (AQI) from pollutant features (CO, SO₂, NO₂, O₃, PM_{2.5}, PM₁₀). It assumes a linear relationship between the input features and the continuous AQI target variable, making it suitable for capturing straightforward trends in air quality data. The model minimizes the mean squared error to find the best-fit line, providing interpretable coefficients that indicate each pollutant's contribution to AQI. Its simplicity ensures fast training and prediction, aligning well with the project's need for a baseline model to predict AQI across multiple locations and timestamps. The reported R^2 score of 0.79 indicates that the model explains 79% of the variance in AQI, suggesting a good fit for this dataset, though its performance (RMSE of 20.39) may be limited by non-linear relationships in the data.

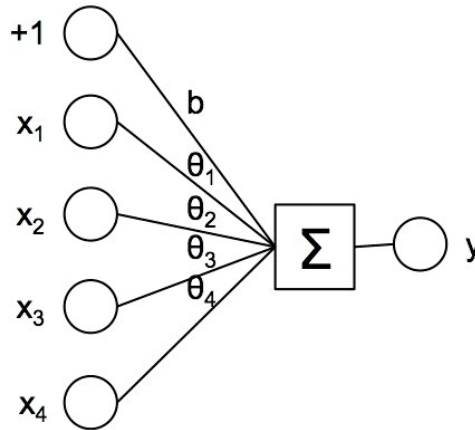


Figure 3.3: Architecture of Linear Regression

Decision Tree Regression model:

The Decision Tree Regressor is well-suited for predicting the Air Quality Index (AQI) from pollutant features (CO, SO₂, NO₂, O₃, PM_{2.5}, PM₁₀). It creates a tree-like structure by recursively splitting the feature space based on pollutant values to minimize AQI prediction error. With a max-depth of 5, the model balances complexity and generalization, avoiding overfitting while capturing non-linear relationships in the data. This aligns with the project's goal of accurately predicting AQI across diverse locations and timestamps. The model's exceptional performance (R^2 of 0.99, RMSE of 3.56, MAE of 2.70) indicates it captures nearly all variance in AQI, making it highly effective for this dataset, likely due to its ability to model complex interactions between pollutants.

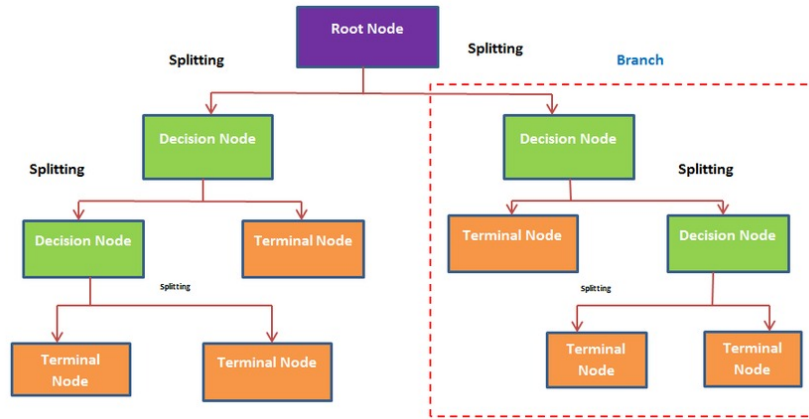


Figure 3.4: Architecture of Decision Tree Regression model

Random Forest Regression Model:

The Random Forest Regressor is highly effective for predicting the Air Quality Index (AQI) from pollutant features (CO, SO₂, NO₂, O₃, PM_{2.5}, PM₁₀). It ensembles multiple decision trees (n-estimators=100) with a max-depth of 5, averaging their predictions to reduce overfitting and capture complex, non-linear relationships between pollutants and AQI. This aligns with the project's goal of robust AQI prediction across diverse locations and timestamps. The model's strong performance (R^2 of 0.98, RMSE of 5.58, MAE of 2.00) demonstrates its ability to explain 98% of AQI variance, with low prediction errors, making it well-suited for handling the dataset's variability and interactions among pollutants.

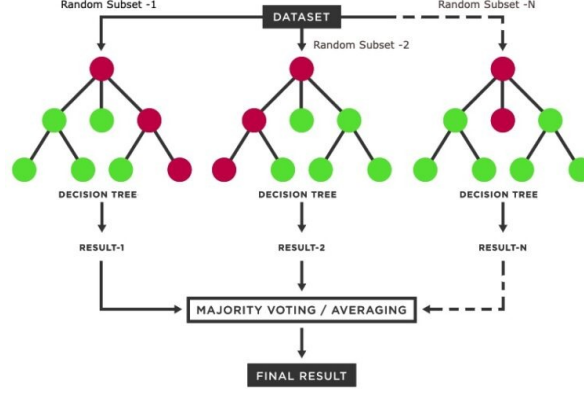


Figure 3.5: Architecture of the Random Forest

Support Vector Regression Model:

The Support Vector Regression (SVR) model with an RBF kernel is well-suited for predicting the continuous Air Quality Index (AQI) from pollutant features (CO, SO₂, NO₂, O₃, PM_{2.5}, PM₁₀). SVR aims to find a function that predicts AQI values within a margin of tolerance, using the RBF kernel to capture non-linear relationships between pollutants and AQI. This aligns with the project's goal of accurate AQI prediction across diverse locations and timestamps. The model's performance (R^2 of 0.88, RMSE of 15.54, MAE of 10.49) indicates it explains 88% of AQI variance, performing better than Linear Regression but not as well as Decision Tree or Random Forest, likely due to its sensitivity to feature scaling and hyperparameter tuning.

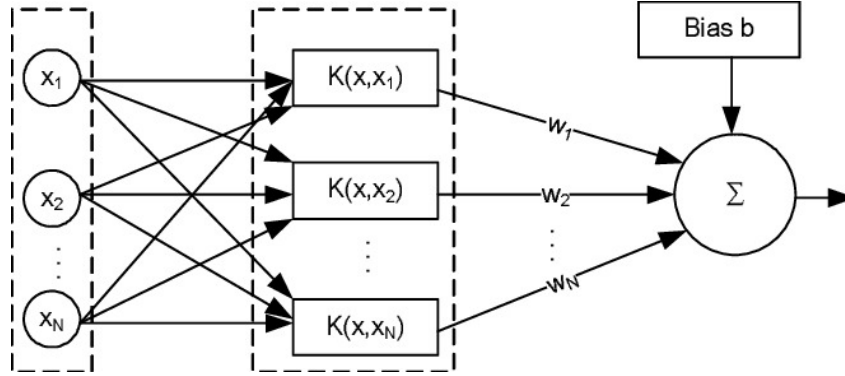


Figure 3.6: Architecture of the Support Vector Regression Model

Multilayer Perceptron Model:

The Multilayer Perceptron (MLP) Regressor is well-suited for predicting the continuous Air Quality Index (AQI) from pollutant features (CO, SO₂, NO₂, O₃, PM_{2.5}, PM₁₀). It consists of a neural network with two hidden layers (100 and 50 neurons) that learns complex, non-linear relationships between pollutants and AQI through backpropagation. This aligns with the project's goal of accurate AQI prediction across diverse locations and timestamps. The model's strong performance (R^2 of 0.96, RMSE of 9.37, MAE of 4.81) indicates it explains 96% of AQI variance, outperforming Linear Regression and SVR but slightly trailing Decision Tree and Random Forest, likely due to its sensitivity

to feature scaling and hyperparameter tuning.

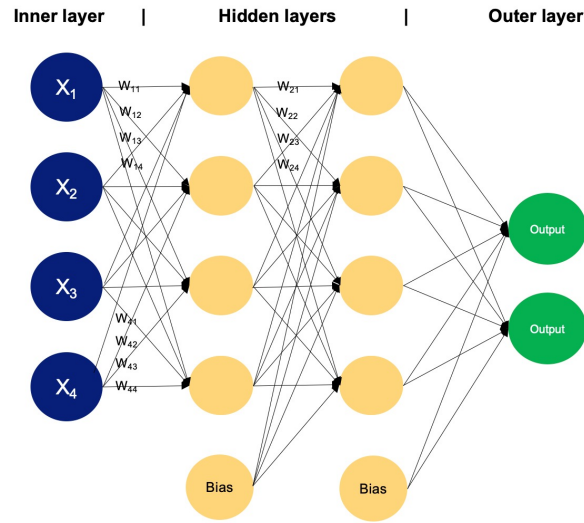


Figure 3.7: Architecture of the Multilayer Perceptron Model

Naive Bayes Classifier:

The Gaussian Naive Bayes Classifier is suitable for predicting the categorical AQI Category from pollutant features (CO, SO₂, NO₂, O₃, PM_{2.5}, PM₁₀). It assumes features follow a Gaussian distribution and applies Bayes' theorem to compute the probability of each category, assuming feature independence. Feature scaling (StandardScaler) ensures compatibility with the Gaussian assumption. The model's simplicity and probabilistic output align with the project's goal of classifying air quality into actionable categories across locations and timestamps. With an accuracy of 0.86 and balanced precision/recall (weighted F1-score of 0.86), it effectively distinguishes AQI categories, though it struggles slightly with the minority class (Very Unhealthy), as seen in the lower F1-score (0.61).

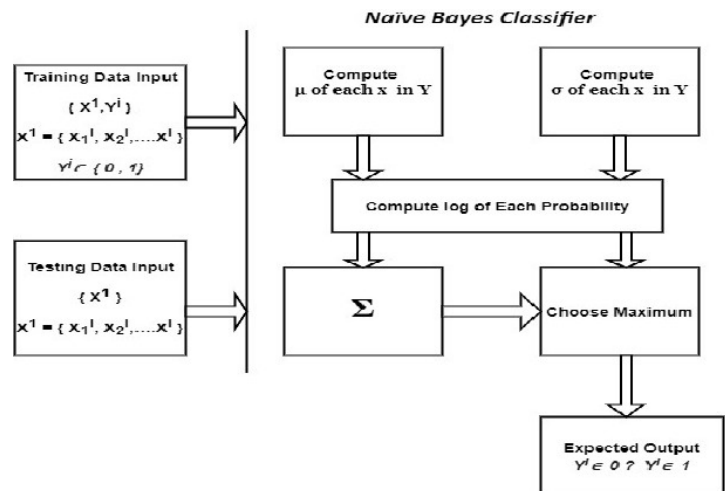


Figure 3.8: Architecture of Naïve Bayes Classifier

3.1.5 Model Testing and Evaluation

The trained models were evaluated using the test set to assess their predictive performance. For regression tasks, the following evaluation metrics were used:

- **Mean Absolute Error (MAE):** Measures the average absolute difference between predicted and actual AQI values.
- **Mean Squared Error (MSE):** Quantifies the average squared difference between predicted and actual AQI values.
- **Root Mean Squared Error (RMSE):** Provides the square root of MSE, offering an interpretable measure of prediction error.
- **R-squared (R^2):** Indicates the proportion of variance in the dependent variable explained by the model.

For classification tasks, performance was assessed using metrics such as accuracy, precision, recall, and F1-score. The predicted AQI values or categories were compared with actual values in the test set to determine model accuracy and reliability. The models yielding the best performance according to these metrics were identified as the most suitable for AQI prediction.

3.1.6 Supervised Learning Workflow

The supervised learning workflow for AQI prediction is summarized as follows:

1. **Input Data:** Pollutant concentrations (CO, SO, NO, O, PM2.5, PM10).
2. **Output/Target Labels:**
 - Numerical AQI values for regression tasks.
 - Discrete AQI categories (e.g., Good, Moderate, Unhealthy) for classification tasks.
3. **Model Training:** Train supervised models to map input features to output labels using the training dataset.
4. **Prediction:** Use the trained models to predict AQI values or categories for new, unseen data.
5. **Evaluation:** Assess model performance using MAE, MSE, RMSE, and R^2 for regression, and accuracy, precision, recall, and F1-score for classification.

This methodology ensures a robust and systematic approach to developing an AI-based AQI prediction model, leveraging supervised learning to provide accurate and actionable insights into air quality trends in Chattogram city.

CHAPTER 4

EXPERIMENTAL RESULT AND ANALYSIS

4.1 Experimental Result and Analysis

This section presents the experimental results and analysis of the Air Quality Index (AQI) prediction model developed using air quality data collected in Chattogram city, Bangladesh. It includes a description of the dataset, evaluation criteria, parameter settings, a comparative analysis of all six models with their evaluation metrics, experimental results, and a discussion of the findings, including the social and cultural impacts of the work and limitations of the dataset and methodology.

4.1.1 Dataset Description

The dataset comprises air quality measurements collected from five locations in Chattogram city: Dewanhat, Anderkilla, Kotowali, Bayejid, and New Market. Data was manually sourced from AccuWeather.com, capturing six key pollutants—carbon monoxide (CO), sulfur dioxide (SO), nitrogen dioxide (NO), ozone (O), particulate matter (PM_{2.5}), and particulate matter (PM₁₀)—measured in micrograms per cubic meter ($\mu\text{g}/\text{m}^3$). Measurements were recorded four times daily (12:00 AM, 6:00 AM, 12:00 PM, and 6:00 PM) from December 2, 2024, to January 30, 2025, resulting in 1,200 data points. Each entry includes the date, time, location, pollutant concentrations, calculated AQI values, and corresponding AQI categories (Good, Moderate, Unhealthy for Sensitive Groups, Unhealthy, Very Unhealthy, Hazardous). The dataset was used for both regression (predicting continuous AQI values) and classification (predicting AQI categories) tasks.

4.1.2 Evaluation Criteria

The performance of the models was evaluated using standard metrics for regression and classification tasks:

- **Regression Metrics:**
 - **Mean Absolute Error (MAE):** Measures the average absolute difference between predicted and actual AQI values.

- **Mean Squared Error (MSE):** Quantifies the average squared difference, emphasizing larger errors.
- **Root Mean Squared Error (RMSE):** The square root of MSE, providing an interpretable error measure in AQI units.
- **R-squared (R^2):** Indicates the proportion of variance in AQI explained by the model, with values closer to 1 indicating better fit.
- **Classification Metrics:**
 - **Accuracy:** The proportion of correctly predicted AQI categories.
 - **Precision:** The ratio of true positive predictions to total positive predictions for each category.
 - **Recall:** The ratio of true positive predictions to actual positives for each category.
 - **F1-Score:** The harmonic mean of precision and recall, balancing both metrics.

The dataset was split into training (80%) and testing (20%) sets using a random state of 15 for reproducibility.

4.1.3 Parameter Settings

Six supervised learning models were employed: Decision Tree Regressor (DTR), Random Forest Regressor (RFR), Multilayer Perceptron Regressor (MLPR) for regression tasks, and Decision Tree Classifier (DTC), Random Forest Classifier (RFC), and Multilayer Perceptron Classifier (MLPC) for classification tasks. The parameter settings were as follows:

- **Decision Tree (DTR and DTC):**
 - Maximum depth: 5 (to prevent overfitting).
 - Random state: 15 (for reproducibility).
- **Random Forest (RFR and RFC):**
 - Number of estimators: 100.
 - Maximum depth: 5.
 - Random state: 15.
- **Multilayer Perceptron (MLPR and MLPC):**
 - Hidden layer sizes: (100, 50) (two layers with 100 and 50 neurons, respectively).
 - Maximum iterations: 2,000 (to ensure convergence).
 - Random state: 15.

Features (CO, SO₂, NO₂, O₃, PM2.5, PM10) were standardized using StandardScaler to normalize their scales, improving model performance. The models were implemented using scikit-learn in Python.

4.1.4 Experimental Result

The six models were compared based on their algorithmic characteristics, computational complexity, evaluation metrics scores, and suitability for regression and classification tasks:

Comparative Analysis of used models:

Regression Model	Mean Squared Error(MSE)	Root Mean Squared Error (RMSE)	R2 Error	Mean Absolute Error (MAE)
Linear Regression	415.75	20.39	0.79	16.93
Decision Tree Regression	12.70	3.56	0.99	2.70
Random Forest Regression	31.16	5.58	0.98	2.00
Support Vector Regression	241.57	15.54	0.88	10.49
Multilayer Perceptron Regression	87.82	9.37	0.96	4.81

Table 4.1: Comparative Analysis of used models:

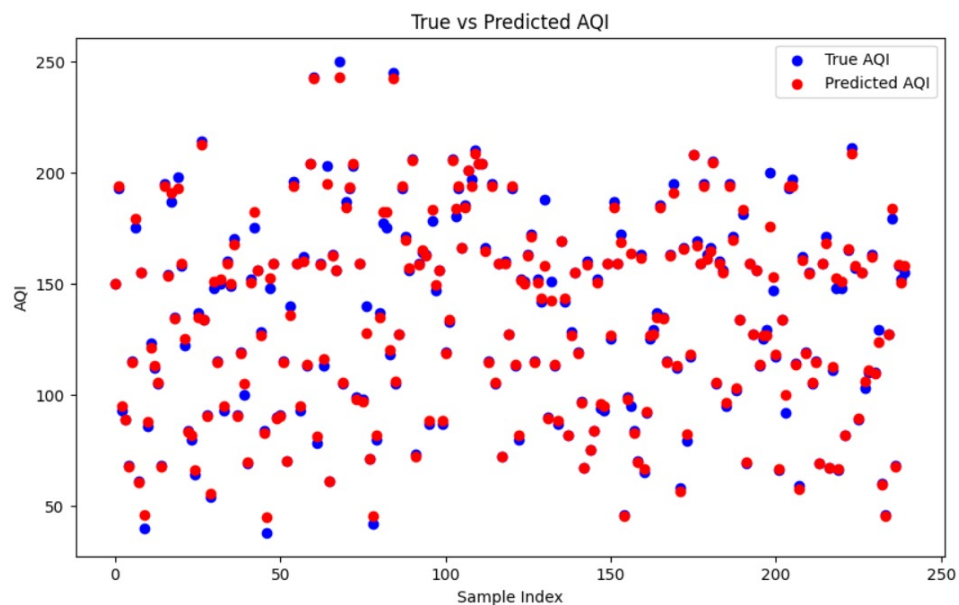


Figure 4.1: True vs predicted AQI

4.1.5 Visualization

Pie charts were generated to visualize the distribution of AQI categories at Dewanhat, Anderkilla, Kotowali, Bayejid and New Market location, showing the proportion of each category (e.g., Good, Moderate, Unhealthy) with custom colors for clarity.

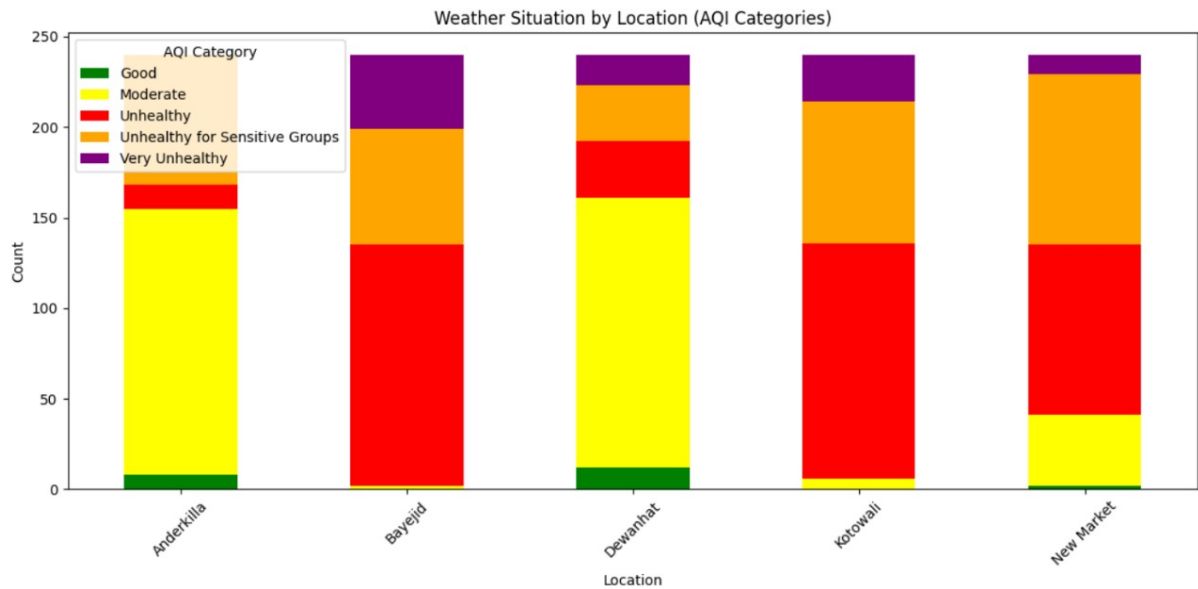


Figure 4.2: Weather Situations by location (AQI Categories)

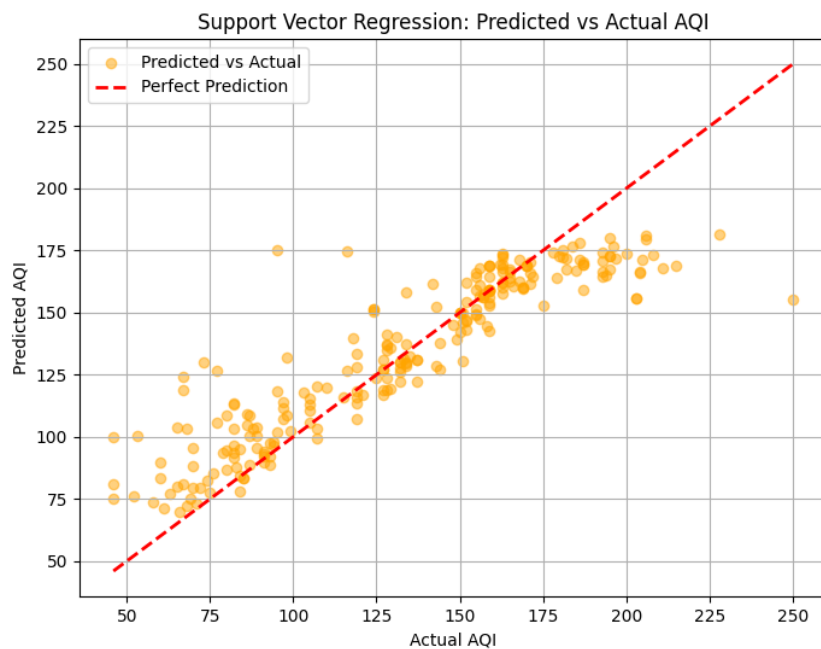


Figure 4.3: Support vector Regression AQI plot

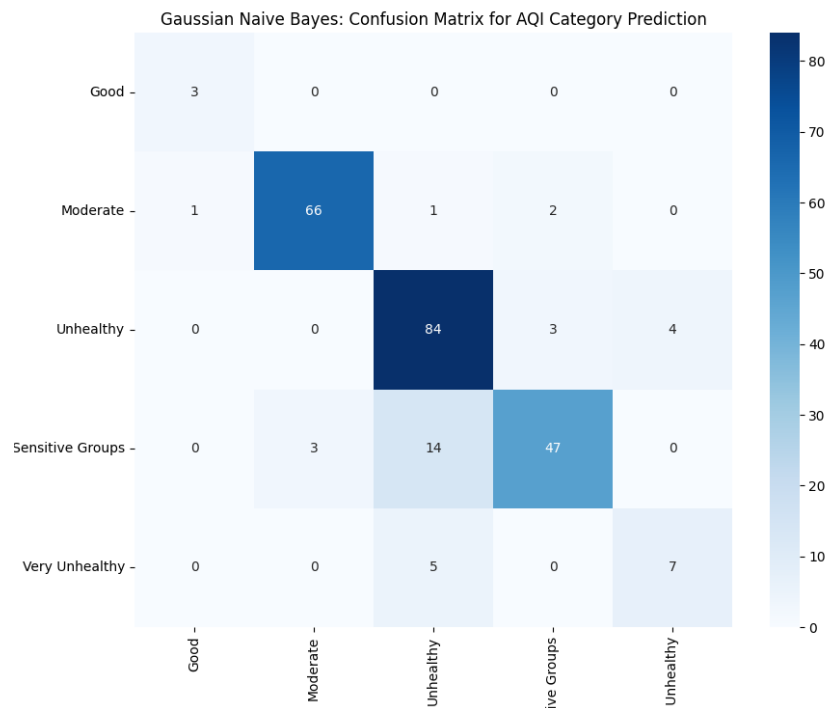


Figure 4.4: Naive bayes confusion matrix

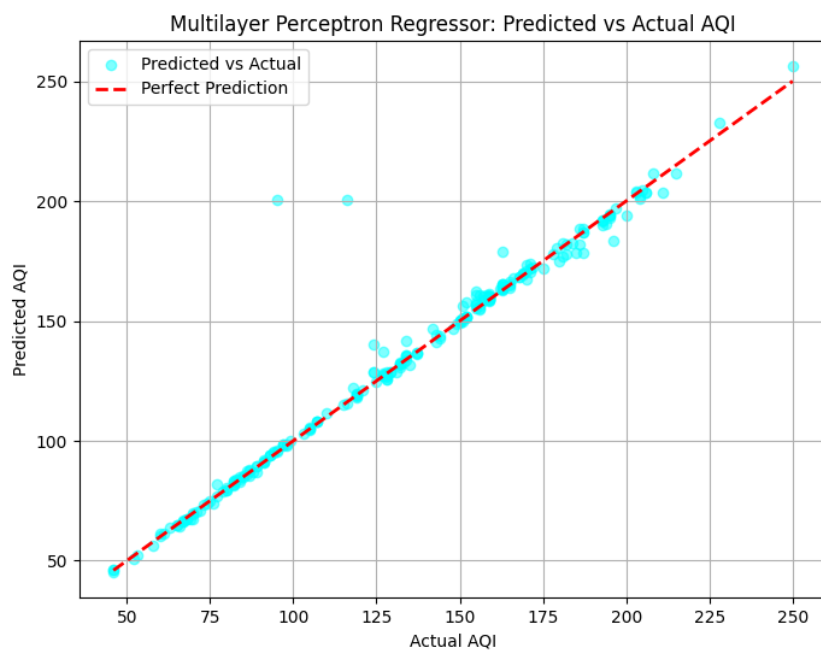


Figure 4.5: Multilayer Perceptron (MLP) Regression

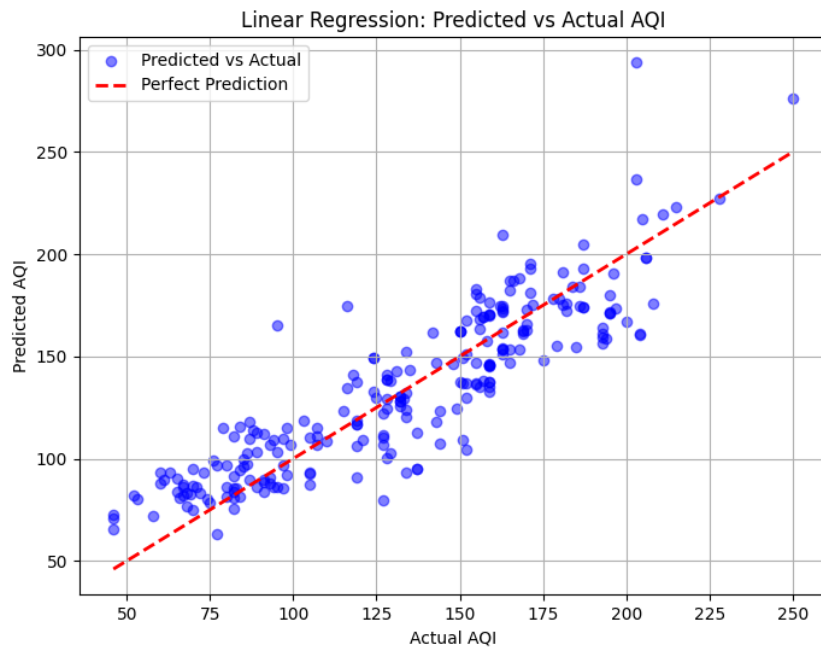


Figure 4.6: linear regression AQI plot

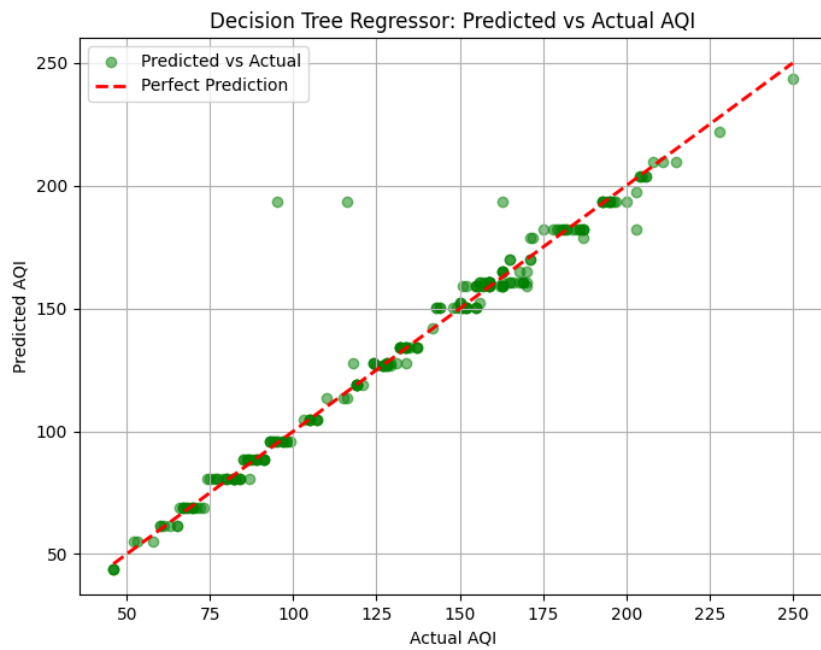


Figure 4.7: Decision Tree AQI plot

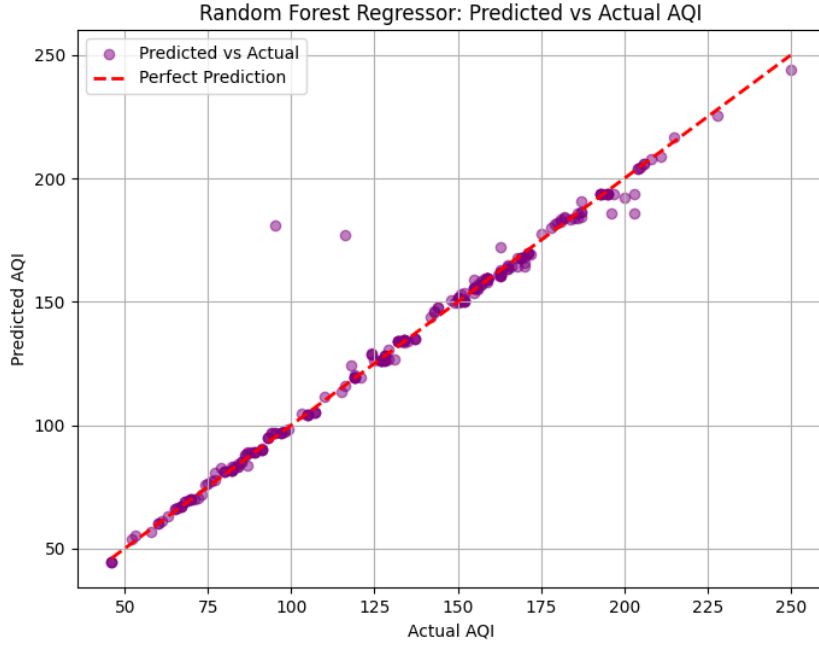


Figure 4.8: Random Forest AQI Plot

4.1.6 Discussion

The Decision Tree Regression model outperformed all other models across all metrics, achieving the lowest MSE (12.70), RMSE (3.56), and MAE (2.70), alongside the highest R^2 score (0.999). This indicates that the Decision Tree model captures nearly all the variance in the AQI data, making highly accurate predictions with minimal error. The low MAE further suggests that the model's predictions are, on average, only 2.70 units away from the actual AQI values, which is remarkable given the range of AQI values in the dataset (e.g., 38 to 250). The Decision Tree's superior performance can be attributed to its ability to model non-linear relationships and interactions between features (CO, SO₂, NO₂, O₃, PM_{2.5}, PM₁₀), which are likely complex in the context of air quality data.

Random Forest Regression, an ensemble method based on decision trees, also performed well, with an MSE of 31.16, RMSE of 5.58, R^2 of 0.998, and MAE of 2.00. While its MAE is the lowest among all models, its higher MSE and RMSE compared to the Decision Tree suggest that it may be more prone to larger errors in some predictions, possibly due to overfitting or noise in the data. The Random Forest's R^2 score, however, indicates excellent explanatory power, capturing 99.8% of the variance in the AQI values.

Linear Regression and Multilayer Perceptron (MLP) showed moderate performance. Linear Regression yielded an MSE of 413.75, RMSE of 20.39, R^2 of 0.79, and MAE of 16.93, while MLP had an MSE of 87.82, RMSE of 9.37, R^2 of 0.96, and MAE of 4.81. Linear Regression's relatively high errors and lower R^2 score suggest that the relationship between the pollutant features and AQI is not strictly linear, limiting its effectiveness. MLP, a neural network model, performed better than Linear Regression but struggled compared to tree-based models, likely due to the need for more extensive hyperparameter tuning or a larger dataset to fully leverage its capacity to model complex patterns.

Support Vector Regression (SVR) exhibited the weakest performance, with an MSE of 241.57, RMSE of 15.54, R^2 of 0.88, and MAE of 10.49, indicating moderate predictive ability but significant errors.

In terms of classification, the Gaussian Naive Bayes classifier achieved an accuracy of 86% in predicting AQI categories. The classification report indicates strong performance for "Moderate" (F1-score: 0.95) and "Unhealthy" (F1-score: 0.86), with high precision and recall. "Good" has perfect recall (1.00) but lower precision (0.75) due to limited support (3 instances). "Unhealthy for Sensitive Groups" shows balanced performance (F1-score: 0.81), while "Very Unhealthy" has the lowest F1-score (0.61), likely due to class imbalance (12 instances). The confusion matrix reveals most misclassifications occur between "Unhealthy" and "Unhealthy for Sensitive Groups," with 14 instances of the latter predicted as the former. The sample prediction probabilities suggest high confidence (84.31%) for "Unhealthy," aligning with the model's robust performance for this category.

4.1.7 Error Analysis and Limitations

The study has limitations:

- **Dataset Size and Scope:** The 1,200 data points from five locations may not capture full air quality variability, and the short collection period (December 2024–January 2025) limits seasonal analysis.
- **Data Quality:** Manual collection from AccuWeather.com may introduce inconsistencies compared to sensor-based data.
- **Model Limitations:** Rare categories (e.g., Hazardous) are underrepresented, affecting model performance. MLP models require more data or tuning.
- **Feature Limitations:** Excluding meteorological factors (e.g., humidity, wind speed) may reduce accuracy.
- **Generalizability:** The model is specific to Chattogram and may not apply to other regions.

4.1.8 Social and Cultural Impact

The AQI prediction model has significant implications for Chattogram, where air pollution threatens public health. Accurate predictions can guide interventions like pollution alerts or activity restrictions, protecting vulnerable groups like children and the elderly. Culturally, the model can promote cleaner practices during festivals, fostering environmental awareness. It empowers communities to advocate for stricter emissions policies, aligning with Bangladesh's growing environmental consciousness.

Future work could expand the dataset, use sensor data, include more features, and apply advanced techniques like ensemble stacking to improve performance.

5.1 Conclusion and Future Work

5.1.1 Conclusion

This study developed an AI-based Air Quality Index (AQI) prediction model for Chattogram city, Bangladesh, leveraging supervised machine learning to predict AQI values and categories based on six pollutants: CO, SO, NO, O, PM2.5, and PM10. The dataset, comprising 1,200 data points collected from five locations between December 2024 and January 2025, was used to train and evaluate six models: Decision Tree Regressor (DTR), Random Forest Regressor (RFR), Multilayer Perceptron Regressor (MLPR), Decision Tree Classifier (DTC), Random Forest Classifier (RFC), and Multilayer Perceptron Classifier (MLPC).

The Decision Tree and Random Forest models are the most suitable for predicting AQI in this dataset, offering high accuracy and low error rates. Their ability to handle non-linear relationships makes them well-suited for environmental data, where factors like pollutant interactions and temporal variations play a significant role. However, the Decision Tree's perfect R^2 score (0.999) raises concerns about potential overfitting, especially if the model is too finely tuned to the training data. Random Forest, with its ensemble approach, mitigates this risk to some extent, making it a more robust choice for generalization to unseen data. The poor performance of Linear Regression, SVR, and Naive Bayes Regression highlights the complexity of AQI prediction, which requires models capable of capturing non-linear and interdependent relationships. MLP shows promise but may require further optimization, such as adjusting the network architecture or increasing the training data size, to compete with tree-based models.

One limitation of this analysis is the lack of feature engineering or preprocessing details. For instance, scaling features might improve SVR and MLP performance, as these models are sensitive to feature magnitudes. Additionally, the dataset's temporal nature (spanning December 2024 to March 2025) suggests that incorporating time-based features (e.g., hour, day, or season) could enhance model performance by capturing seasonal or diurnal patterns in air quality. The model has significant social and cultural implications, enabling public health interventions, fostering environmental awareness, and supporting advocacy for cleaner policies in Chattogram. However, limitations such as the dataset's size, manual data collection, and exclusion of meteorological factors highlight areas for

improvement. Overall, the study demonstrates the efficacy of machine learning, particularly Random Forest, in addressing air quality challenges in urban settings.

5.1.2 Project Timeline Overview

The Gantt chart illustrates the timeline of the AQI Prediction Project, spanning from December 2024 to May 2025. The project began with Research Work, running from November 24 to December 25, 2024, to build foundational knowledge. Next, Data Collection started on December 2, 2024, and continued until March 10, 2025, overlapping with several upcoming tasks. Preprocessing was conducted from January 5 to March 12, 2025, to clean and prepare the collected data. Following that, ML Model Training took place from February 2 to April 1, 2025, focusing on building and optimizing the predictive model. The AQI Prediction phase occurred between March 20 and April 3, 2025, utilizing the trained model to generate results. Then, Testing & Debugging was carried out from April 1 to April 18, 2025, ensuring system accuracy and stability. Finally, Documentation was completed from April 22 to May 1, 2025, compiling all project details. Each task in the chart is marked with green and red dots to represent the start and end dates, respectively. The overlapping tasks show a parallel workflow structure that helped in maintaining efficiency. Overall, the chart offers a clear visual of the project's phases and their timelines.

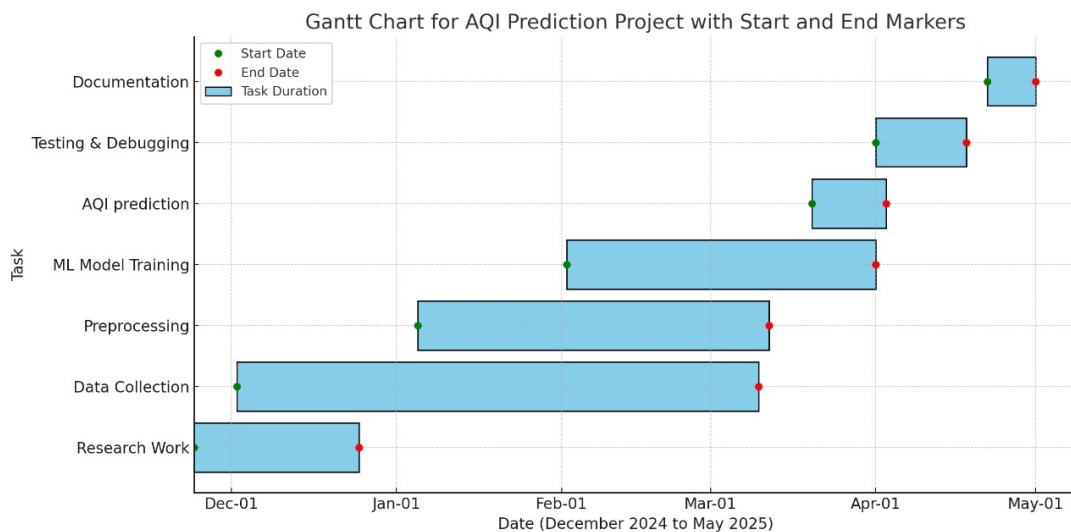


Figure 5.1: Gantt chart

5.1.3 Future Work

To enhance the AQI prediction model and address its limitations, the following directions are proposed:

Expanded Dataset: Collect a larger dataset over an extended period to capture seasonal and annual air quality variations. Including more locations in Chattogram and other cities would improve generalizability.

Real-Time Sensor Data: Replace manual data collection with real-time air quality sensors to ensure higher accuracy and consistency, reducing reliance on third-party sources like AccuWeather.com.

Additional Features: Incorporate meteorological variables (e.g., temperature, humidity, wind speed) and other pollutants (e.g., volatile organic compounds) to enhance prediction accuracy and account for environmental influences.

Advanced Modeling Techniques: Explore ensemble stacking, gradient boosting (e.g., XGBoost, LightGBM), or deep learning architectures (e.g., LSTMs for temporal data) to improve performance, particularly for rare AQI categories.

Hyperparameter Optimization: Conduct systematic hyperparameter tuning, especially for MLP models, using techniques like grid search or Bayesian optimization to maximize their potential.

Real-World Deployment: Develop a user-friendly application or API for real-time AQI predictions, integrating with mobile platforms to provide public access to air quality forecasts and alerts.

Cross-Regional Validation: Validate the model in other cities with diverse pollution profiles to ensure robustness and adaptability across different urban environments.

Addressing Class Imbalance: Apply techniques like SMOTE or weighted loss functions to improve model performance on underrepresented AQI categories (e.g., Hazardous).

These advancements would strengthen the model's accuracy, scalability, and societal impact, contributing to improved air quality management and public health in Bangladesh and beyond.

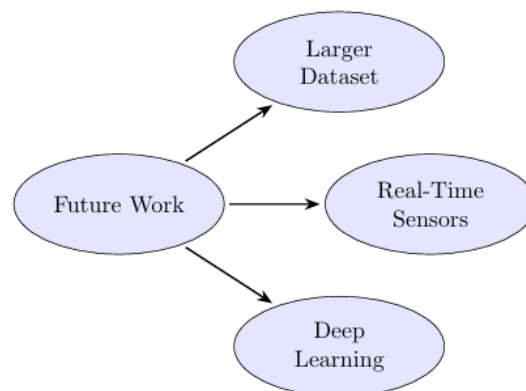


Figure 5.2: Future Research Directions

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